

10(**short**) Figure 9 shows a smooth rod of length L rotating anti-clockwise in a horizontal plane about O at a constant angular velocity Ω . A particle P, which is free to slide along the rod, is attached to a light inextensible wire of length $L/2$ whose other end is connected to O. At the instant shown in Fig. 9, the wire snaps and the particle P slides towards the end of the rod.

- (a) Show that the radial position r of P while P is still in contact with the rod, expressed as a function of time t after the wire snapped, is given by $r = A \cosh bt$ and determine A and b in terms of L and Ω . [6]
- (b) Derive an expression for the time taken for P to reach the end of the rod. [4]

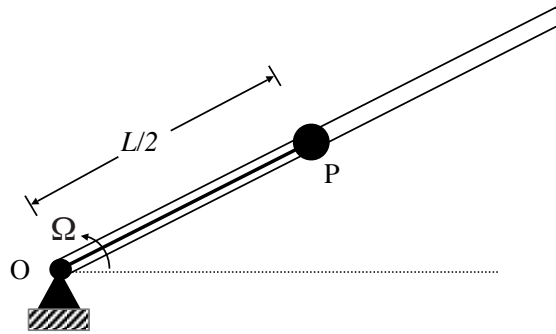


Fig. 9